Corporate Bankruptcy Prediction using Accounting Ratios

Situation and Complication

Just as doctors check blood pressure and pulse rate as vital indicators of health of a patient, business analysts also scour the financial statements of a corporation to monitor its financial health. A primary issue in financial analysis, then is, how predictable is the health of a company? Using a variety of financial indicators that are available in this dataset of 132 firms (from the manufacturing and retail sectors), 66 of which filed for bankruptcy, your task is to develop a bankruptcy prediction model and, more importantly, to explain how that model makes bankruptcy predictions.

Data

The dataset for this case is “bankruptcy.csv”. The dataset contains the following fields:

- **D**: D=0 for failed firms, D=1 for healthy firms (this is the target variable).

And the following financial ratios are available in the dataset:

- **R1**: CASH/CURDEBT
- **R2**: CASH/SALES
- **R3**: CASH/ASSETS
- **R4**: CASH/DEBTS
- **R5**: CFF0/SALES
- **R6**: CFFO/ASSETS
- **R7**: CFFO/DEBTS
- **R8**: COGS/INV

- **R9**: CURASS/CURDEBT
- **R10**: CURASS/SALES
- **R11**: CURRASS/ASSETS
- **R12**: CURDEBT/DEBTS
- **R13**: INC/SALES
- **R14**: INC/ASSETS
- **R15**: INC/DEBTS
- **R16**: UBCDEP/SALES
- **R17**: INCDEP/ASSETS
- **R18**: INCDEP/DEBTS
- **R19**: SALES/REC
- **R20**: SALES/ASSETS
- **R21**: ASSETS/DEBTS
- **R22**: WCFO/SALES
- **R23**: WCFO/ASSETS
- **R24**: WCFO/DEBTS

Here are the explanations of abbreviations in financial ratios:

- ASSETS: Total assets
- CASH: Cash
- CFFO: Cash flow from operations
- COGS: Cost of goods sold
- CURASS: Current assets
- CURDEBT: Current debt
- DEBTS: Total debt
- INC: Income
- INCDEP: Income plus depreciation

- INV: Inventory
- REC: Receivables
- SALES: Sales
- WCFO: Working capital from operations

Key Questions

1. Follow the comments in “HW2 script.ipynb” to complete the bankruptcy prediction model training / evaluation / interpretation tasks.
2. Are the global interpretations extracted from different interpretation methods consistent with each other or not? If not, how do they differ?
3. Suppose you are a data scientist working for a company that uses a bankruptcy prediction model as part of its business (e.g., a management consulting firm relying on predictions of clients’ bankruptcy likelihood to tailor its strategic recommendations). How would you present the results of global interpretations to a group of executives in the company? Please note that this question is not simply asking you to pick one predictive model you have built in question #1 and report its global interpretations. You need to formulate your answer based on your findings in the above question #2.
4. Now putting yourself in the shoes of a consultant in this company, who works with clients in mitigating bankruptcy risks. Based on your answers to question #3, what business recommendations would you give to your clients?